

Continuity and Differentiability: Formula Sheet

Class 12 Mathematics – Chapter 5 Complete Reference & Formulas

1 Continuity of Real Functions

- **Continuity at a Point $x = c$:** A function $f(x)$ is continuous at $x = c$ if:

$$\lim_{x \rightarrow c} f(x) = f(c) \iff \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$$

- **Algebra of Continuous Functions (Theorem 1):** If f and g are continuous at $x = c$, then:
 - $(f \pm g)(x)$ is continuous at $x = c$.
 - $(f \cdot g)(x)$ and $(k \cdot f)(x)$ are continuous at $x = c$.
 - $\left(\frac{f}{g}\right)(x)$ is continuous at $x = c$, provided $g(c) \neq 0$.
- **Continuity of Composite Functions (Theorem 2):** If g is continuous at c and f is continuous at $g(c)$, then $(f \circ g)(x) = f(g(x))$ is continuous at c .

2 Differentiability & Fundamental Theorem

- **Derivative at a Point $x = c$ (First Principle):**

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

Left Hand Derivative (LHD) = Right Hand Derivative (RHD):

$$\lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$$

- **Differentiability Implies Continuity (Theorem 3):**

$$f(x) \text{ is differentiable at } x = c \implies f(x) \text{ is continuous at } x = c$$

Converse is NOT true (e.g., $f(x) = |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$).

3 Algebra of Derivatives & Standard Rules

- 1. Linearity Rule:

$$\frac{d}{dx}[u(x) \pm v(x)] = \frac{du}{dx} \pm \frac{dv}{dx}$$

- 2. Product Rule (Leibniz Rule):

$$\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$$

For three functions: $\frac{d}{dx}(u \cdot v \cdot w) = \frac{du}{dx}vw + u\frac{dv}{dx}w + uv\frac{dw}{dx}$.

- 3. Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad (v \neq 0)$$

- 4. Chain Rule (Composite Functions): If $y = f(t)$ and $t = g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

4 Standard Derivatives Table

Function $f(x)$	Derivative $f'(x)$	Domain / Restriction
x^n	$n x^{n-1}$	$n \in \mathbb{R}$
$\sin x$	$\cos x$	$\mathbb{R}$
$\cos x$	$-\sin x$	$\mathbb{R}$
$\tan x$	$\sec^2 x$	$x \neq (2n+1)\frac{\pi}{2}$
$\cot x$	$-\operatorname{cosec}^2 x$	$x \neq n\pi$
$\sec x$	$\sec x \tan x$	$x \neq (2n+1)\frac{\pi}{2}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$	$x \neq n\pi$
e^x	e^x	$\mathbb{R}$
a^x	$a^x \ln a$	$a > 0, a \neq 1$
$\log_e x$ ($\ln x$)	$\frac{1}{x}$	$x > 0$
$\log_a x$	$\frac{1}{x \ln a}$	$x > 0, a > 0, a \neq 1$

5 Derivatives of Inverse Trigonometric Functions

Inverse Function	Derivative $\frac{dy}{dx}$	Domain of Derivative
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\mathbb{R}$
$\cot^{-1} x$	$-\frac{1}{1+x^2}$	$\mathbb{R}$
$\sec^{-1} x$	$\frac{1}{ x \sqrt{x^2-1}}$	$ x > 1$
$\operatorname{cosec}^{-1} x$	$-\frac{1}{ x \sqrt{x^2-1}}$	$ x > 1$

6 Logarithmic Differentiation

- **General Formulation for Variable Powers** ($y = [u(x)]^{v(x)}$): Take natural logarithm of both sides:

$$\ln y = v(x) \cdot \ln[u(x)]$$

$$\frac{1}{y} \frac{dy}{dx} = v(x) \cdot \frac{1}{u(x)} \cdot u'(x) + v'(x) \cdot \ln[u(x)]$$

$$\frac{dy}{dx} = y \left[\frac{v(x)}{u(x)} \cdot u'(x) + v'(x) \cdot \ln[u(x)] \right]$$

- **Fundamental Logarithm Identities Used in Calculus:**

$$\log_a p = \frac{\log_b p}{\log_b a} = \frac{\ln p}{\ln a}$$

$$\ln(p \cdot q) = \ln p + \ln q$$

$$\ln\left(\frac{p}{q}\right) = \ln p - \ln q$$

$$\ln(p^n) = n \ln p, \quad e^{\ln x} = x \quad (x > 0)$$

7 Parametric & Second Order Derivatives

- **Parametric Differentiation** ($x = f(t)$, $y = g(t)$):

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)} \quad \left(\text{provided } \frac{dx}{dt} \neq 0 \right)$$

- **Second Order Derivative of Explicit Function** $y = f(x)$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = f''(x) = y'' = y_2$$

- **Second Order Derivative for Parametric Equations:**

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$